

Department of computer science and Engineering -1 -- KLAZIZ K L Deemed to be University Course Handout 2025-2026, Term3	
Course Title	: COMPUTATIONAL FOUNDATIONS FOR ARTIFICIAL INTELLIGENCE
Course Code	: 25SC1306E
L-T-P-S Structure	: 2-0-0-4
Pre-requisite	:
Credits	: 3
Course Coordinator	: Sudarshan Yerragunta
Team of Instructors	:
Teaching Associates	:
Syllabus :	CO1: Agent model (PEAS), environment types; problem formulation (state/action/transition/cost); knowledge representation types: graphs, trees, rule sets, constraints; Python essentials for AI algorithms: functions, recursion, classes for state, dataclasses, typing hints, complexity-aware coding; core DS in Python: dict/set/list, heaps/priority queue; trace logging & step-by-step reasoning; testing small algorithm units. CO2: BFS/DFS/UCS; A* and Greedy; heuristic design and evaluation; admissibility/consistency intuition; closed/open sets; tie-breaking strategies; memory-bounded variants concept (IDA* idea only); empirical profiling (node expansions, runtime, peak memory); search on graphs & implicit state spaces (puzzles). CO3: CSP modeling; backtracking; constraint propagation (forward checking, arc consistency intuition); heuristics MRV/degree/LCV; local search for CSP (min-conflicts); scheduling/timetabling; SAT-intuition (very light) as modeling tool; explainability: why a constraint failed. Co4: Utility functions; minimax; alpha-beta pruning; evaluation functions; depth limits; iterative deepening concept; stochastic decisions overview (expectimax concept); bounded rationality; policy selection; multi-agent reasoning basics. CO5: Probability review only as required; Bayes rule; Bayesian networks (structure, CPT); inference by variable elimination (worked examples), belief propagation intuition; sampling inference (rejection/likelihood weighting concept); Markov chains & HMM intuition for tracking; sensor fusion reasoning cases; uncertainty-aware decisions (expected utility concept). CO6: Combine search + CSP + probabilistic inference + decision logic; hybrid architectures (rule + search + probabilistic); explainable reasoning traces; performance engineering and failure analysis; ethics/limitations (technical: bias in heuristics, uncertainty miscalibration); preparing readiness for ML/DL/NLP/GenAI courses.
Text Books :	1. S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th ed. Pearson, 2021. 2. D. Poole and A. Mackworth, Artificial Intelligence: Foundations of Computational Agents, 2nd ed. Cambridge, U.K.: Cambridge University Press, 2017.
Reference Books :	1. E. Rich and K. Knight, Artificial Intelligence, 2nd ed. New York, NY, USA: McGraw-Hill, 1991. 2. Aggarwal, Flask Framework Cookbook. Birmingham, U.K.: Packt Publishing, 2018. 3. W. J. Chun, Core Python Programming, 3rd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2012.
Course Rationale :	This course builds a foundation in Artificial Intelligence by combining theory with practical implementation, guided by Artificial Intelligence: A Modern Approach. The main goal is to enable students to design, implement, and evaluate intelligent systems that can solve problems, make decisions, and handle uncertainty effectively.
Course Objectives :	The objective of this course is to provide a comprehensive understanding of artificial intelligence by introducing the fundamentals of intelligent agents, environments, and problem formulation, along with knowledge representation techniques such as graphs, trees, rules, and constraints, supported by essential Python programming skills for AI development; to develop the ability to design and analyze both uninformed and informed search algorithms and effective

heuristics for problem solving; to model and solve real-world problems using constraint satisfaction techniques and heuristics; to understand decision-making in competitive and multi-agent environments through game-playing algorithms; to apply probabilistic reasoning and inference methods for handling uncertainty; and to integrate these approaches to build efficient, explainable, and ethical AI systems

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Formulate real-world tasks as AI problems (state space, actions, transitions, goals, costs, constraints) and implement representations using Python + core data structures (lists/dicts/classes) with traceable reasoning.	PO2,PO1	4
CO2	Implement and analyze graph/state-space search algorithms (BFS/DFS/UCS/A*/Greedy) and design heuristics with admissibility/consistency reasoning; empirically compare time/space trade-offs.	PO1,PO2,PO4	5
CO3	Model and solve constraint satisfaction problems (CSP) using backtracking, propagation, and heuristics (MRV/degree/LCV), and evaluate solution quality for realistic scheduling/resource tasks	PO1,PO3,PO4	5
CO4	Design decision-making agents for sequential/adversarial settings using minimax/alpha-beta and utility- based policy selection; analyze performance and failure modes under bounded computation.	PO1,PO4,PO2	5
CO5	Apply reasoning under uncertainty using Bayesian networks and Markov models and perform inference using algorithmic methods (variable elimination/message passing intuition, sampling basics) for diagnosis/tracking cases. (No statistics overlap)	PO1,PO2	5
CO6	Build integrated AI reasoning pipelines in Python combining representation + search/CSP + uncertainty + decision logic, and produce explainable outputs (reasoning traces, constraint proofs, inference steps).	PO3,PO4,PO5,PO10	5

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-2	COI-3	COI-4	COI-5
CO1	4	Btl-2 Explain and describe state-space representation and problem components	Btl-3 Build state space transitions models and cost functions	Btl-4 Examine step-wise state expansion and Validate formulation against goals and constraints	
CO2	5	Btl-2 Illustrate uninformed and informed search strategies and heuristics	Btl-3 Develop heuristic properties (admissibility, consistency)	Btl-4 Inspect the Measure performance (time, memory, node expansions)	Btl-5 Estimate the algorithm efficiency across problem instances

Outcome No.	Highest BTL	COI-2	COI-3	COI-4	COI-5
CO3	5	Btl-2 Describe CSP components (variables, domains, constraints) and solving approach.	Btl-3 Construct MRV, Degree, LCV heuristics	Btl-4 Analyze to Execute CSP on real-world problems	Btl-5 Determine the Assess solution quality and constraint satisfaction
CO4	5	Btl-2 Interpret minimax, alpha-beta pruning, and utility-based decisions	Btl-3 Apply evaluation functions and depth limits	Btl-4 Analyze pruning efficiency and performance	Btl-5 Evaluate agent decisions in adversarial settings
CO5	5	Btl-2 Explain Bayesian networks, Bayes rule, and Markov models	Btl-3 Solve probabilistic inference	Btl-4 Interpret belief updates based on evidence	Btl-5 Evaluate reasoning to diagnosis/tracking problems
CO6	5	Btl-2 Describe integration of AI components in a pipeline	Btl-3 Make use of explainability via reasoning traces	Btl-4 Categorize Test system with structured cases	Btl-5 Consider the Present and justify system outputs

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
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Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-2	Introduction to AI, Agent Model, problem Formulation (state/action/transition/cost)	TB1: [CH1: PAGE NO:5]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEM-EXAM1,SEP,Skill In-Sem Exam
2	CO1	COI-3	knowledge representation types: graphs, trees, rule sets, constraints	TB1: [CH2:P.NO:35]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEM-EXAM1,SEP,Skill In-Sem Exam
3	CO1	COI-3	Python essentials for AI algorithms: functions, recursion, classes for state, data classes, typing hints, complexity-aware coding	RB3: [CH1:P.NO:5]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEM-EXAM1,SEP,Skill In-Sem Exam
4	CO1	COI-4	core DS in Python: dict /set/list, heaps/priority queue; trace logging & step-by-step reasoning; testing small algorithm units.	RB3: [CH3:P.NO:40]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,LCE,SEM-EXAM1,SEP,Skill In-Sem Exam

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
5	CO2	COI-2	Uninformed & Informed Search (Algorithmic Core): BFS/DFS/UCS; Uninformed & Informed Search (Algorithmic Core): BFS/DFS/UCS;	TB1: [CH3: PAGE NO: 64]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEM-EXAM1,SEP,Skill In-Sem Exam
6	CO2	COI-3	A* and Greedy; heuristic design and evaluation;	TB1: [CH3: PAGE NO:90]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEM-EXAM1,SEP,Skill In-Sem Exam
7	CO2	COI-4	admissibility/consistency intuition; closed/open sets; tie-breaking strategies; memory-bounded variants concept (IDA)	TB1: [CH4: PAGE NO:120]	Chalk,LTC,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEM-EXAM1,SEP,Skill In-Sem Exam
8	CO2	COI-5	empirical profiling (node expansions, runtime, peak memory); search on graphs & implicit state spaces (puzzles).	TB1: [CH4: PAGE NO:131]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEM-EXAM1,SEP,Skill In-Sem
9	CO3	COI-2	Constraint Satisfaction & Combinatorial Reasoning: CSP modeling; backtracking; constraint propagation (forward checking, arc consistency intuition);	TB1: [CH6: PAGE NO:202]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEM-EXAM2,SEP
10	CO3	COI-3	heuristics MRV/degree/LCV; local search for CSP (min-conflicts);	TB1: [CH6: PAGE NO:214]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEM-EXAM2,SEP
11	CO3	COI-4	Heuristics (MRV, LCV), Scheduling/timetabling;	TB1: [CH6: PAGE NO:220]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEM-EXAM2,SEP
12	CO3	COI-5	SAT-intuition (very light) as modeling tool; explainability: why a constraint failed.	TB1: [CH6: PAGE NO:227]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEM-EXAM2,SEP
13	CO4	COI-2	Decision Making & Adversarial Search: Utility functions; minimax	TB1: [CH5: PAGE NO:161]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEM-EXAM2
14	CO4	COI-3	stochastic decisions	TB1: [CH5:	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
			overview (expectimax concept);	PAGE NO:163]		Semester Exam,SEM-EXAM2,SEP
15	CO4	COI-4	stochastic decisions overview (expectimax concept	TB1: [CH5: PAGE NO:167]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEM-EXAM2,SEP
16	CO4	COI-5	bounded rationality; policy selection; multi-agent reasoning basics.	TB1: [CH5: PAGE NO:170]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEM-EXAM2,SEP
17	CO5	COI-2	Reasoning Under Uncertainty (Inference, not Statistics): Probability review only as required; Bayes rule; Bayesian networks.	TB1: [CH14: PAGE NO:511]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEP
18	CO5	COI-3	inference by variable elimination (worked examples), belief propagation intuition; sampling inference (rejection/likelihood weighting);	TB1: [CH14: PAGE NO:552]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEP
19	CO5	COI-4	Markov chains & HMM intuition for tracking; sensor fusion reasoning cases;	TB1: [CH14: PAGE NO:554]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEP
20	CO5	COI-5	uncertainty-aware decisions (utility concept).	TB1: [CH17: PAGE NO:666]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEP
21	CO6	COI-2	Build integrated AI reasoning pipelines in Python combining representation.	TB1: [CH17: PAGE NO:561]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEP
22	CO6	COI-3	search/CSP, uncertainty, decision logic	TB1: [CH16: PAGE NO:527]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEP
23	CO6	COI-4	Produce explainable AI outputs	TB1: [CH19: PAGE NO:651]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEP
24	CO6	COI-5	Reasoning traces, constraint proofs, inference steps.	TB1: [CH9: PAGE NO:303]	Chalk,PPT,Talk	ALM,Continuous Evaluation -Project,End Semester Exam,SEP

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	4	PPT	--- NOT APPLICABLE ---
20	An intelligent agent for robot navigation is designed with a structure consisting of sensors to perceive the environment, a processing unit (agent program) to make decisions, and actuators to move the robot toward the goal while avoiding obstacles;	4	PPT	--- NOT APPLICABLE ---
20	The concept of the PEAS model is used to design a robot navigation agent by defining Performance (reach the goal safely and quickly), Environment (grid with obstacles), Actuators (wheels and motors), and Sensors (camera and distance sensors) to describe how the agent interacts with its surroundings.	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	4	Talk	--- NOT APPLICABLE ---
20	Design a robot navigation environment model using a grid map containing obstacles and dynamic elements, then classify the environment based on AI environment types. The project outcome will include a written classification report and environment diagram, demonstrating understanding of real-world AI system design.	4	PPT	--- NOT APPLICABLE ---
20	Graph-based knowledge representation uses nodes to represent entities and edges to represent relationships, while rule-based knowledge representation uses IF-THEN rules to define logical conditions and actions for decision-making in Artificial Intelligence systems	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap	1	Talk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
20	Design a grid navigation problem by formulating an AI problem using state-space representation, where each grid cell represents a state, movements represent actions, and transitions define paths from the start state to the goal state. Design a grid with obstacles, identify initial state, goal state, actions, and path costs, and present a state-space diagram showing all possible navigation paths.	4	PPT	One minute paper
20	State-space representation by identifying states, actions, and transitions, and presenting their navigation diagram showing possible solution paths.	4	Talk	--- NOT APPLICABLE ---
5	Conclusion and Summary Talk	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap	1	Talk	--- NOT APPLICABLE ---
20	Creating a grid navigation model where lists store possible states, dictionaries maintain node-to-cost mappings, and sets track visited nodes during search operations which help in efficient path tracking.	4	PPT	One minute paper
20	Lists, dictionaries, and sets in Python are important data structures used to store and organize data efficiently in Artificial Intelligence applications. In AI problems, lists are used to store states or paths, dictionaries are used to maintain mappings between nodes and values, and sets are used to track visited nodes and avoid repeated processing.	4	Talk	--- NOT APPLICABLE ---
5	Conclusion and Summary	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap	1	Talk	--- NOT APPLICABLE ---
20	Design a robot navigation grid by defining the state space, initial state, goal state, actions, and movement costs, then apply BFS, DFS, and UCS algorithms to explore possible paths. Compare the algorithms by recording path length, total cost, and explored nodes, and present a comparison table showing which algorithm produces the shortest and most efficient path.	5	Talk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
20	Uninformed and cost-based search algorithms such as Breadth-First Search (BFS), Depth-First Search (DFS), and Uniform Cost Search (UCS) are used to explore the state space of a mobile robot navigation problem, where states represent robot positions, the initial state is the starting location, the goal state is the destination, and actions include movements such as up, down, left, and right.	5	PPT	Brain storming session
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap of Search Strategies (uninformed vs informed search)	1	Talk	--- NOT APPLICABLE ---
20	Design a grid-based robot navigation problem using Greedy Best-First Search and A* algorithms selecting nodes based on heuristic values and comparing path results between the two algorithms. They will also calculate Manhattan and Euclidean distances to evaluate heuristic performance and analyze how heuristic choice affects path length and efficiency.	5	Talk	--- NOT APPLICABLE ---
20	Greedy Best-First Search selects nodes using heuristic values to move toward the goal quickly but may fail to find the optimal path, while the A* algorithm combines path cost and heuristic values to find an efficient and optimal path in robot navigation. Heuristic design, such as Manhattan and Euclidean distances, must satisfy properties like admissibility and consistency to ensure correct pathfinding and improved search performance.	5	Talk	Case Study
5	Conclusion and Summary: Comparison of Greedy vs A*, importance of heuristic choice in single-agent systems	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap of A* and Greedy in Robot Navigation	1	Talk	--- NOT APPLICABLE ---
20	Design a navigation grid with obstacles and implement A* search using admissible heuristics, managing open and closed node lists and applying tie-breaking rules during equal-cost scenarios. Students will extend the project by simulating IDA* navigation for memory-constrained robots and presenting PPT (PBL based) Talk performance	1	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
	comparisons showing trade-offs between memory usage and computational effort.			
20	In A* robot navigation, heuristic properties such as admissibility (never overestimating cost) and consistency (maintaining logical cost progression) ensure optimal pathfinding, while open and closed sets manage frontier and explored nodes efficiently during grid traversal with obstacles. and memory-bounded methods like Iterative Deepening A* (IDA*) allow robots with limited memory to find solutions by balancing memory usage and recomputation.	5	PPT	--- NOT APPLICABLE ---
5	Conclusion & Summary: Linking heuristic quality, memory use, and optimal navigation performance	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap (state space, search basics)	1	Talk	--- NOT APPLICABLE ---
20	students will design experiments to run search algorithms on both grid navigation environments and puzzle-based implicit state spaces, systematically collecting performance data. Students will analyze results through charts and comparison reports, demonstrating how empirical profiling helps select the most efficient algorithm for different AI problem types.	5	PPT	--- NOT APPLICABLE ---
20	Empirical profiling in search algorithms involves measuring performance metrics such as node expansions, runtime, and peak memory usage to evaluate the efficiency of search methods during execution. Search on graphs represents problems like grid-based robot navigation, while search in implicit state spaces, such as puzzles, generates states dynamically without explicitly storing all possible states, enabling comparison of algorithm performance across different problem types.	5	Talk	--- NOT APPLICABLE ---
5	Conclusion & Summary (performance trade-offs)	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	Model a robot navigation grid as a CSP by defining valid positions, movement constraints, and obstacle restrictions,	5	PPT	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
	then apply backtracking search to find feasible paths from start to goal. They will use forward checking and arc consistency to reduce invalid moves and compare navigation efficiency with and without constraint propagation.			---
20	In robot navigation, the path-planning problem can be modeled as a Constraint Satisfaction Problem (CSP) where grid positions act as variables, possible movements form domains, and constraints ensure the robot avoids obstacles and follows valid paths. Backtracking explores possible movement sequences, while constraint propagation techniques such as forward checking and arc consistency help eliminate invalid paths early and improve navigation efficiency.	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	Discussion on CSP in Multi-Robot Navigation (modeling multiple robots' paths as CSP: variables = robot positions, domains = moves, constraints = collision avoidance, resource sharing, sensor noise). Backtracking search for multi-robot path planning (illustration with grid world navigation problem under uncertainty; robots coordinating moves)	5	PPT	--- NOT APPLICABLE ---
20	Constraint propagation: Forward checking and Arc Consistency intuition (robots avoid dead-ends, collisions, and inconsistent joint paths under uncertain environments)	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	Discussion on heuristics in CSP for multi-robot navigation (why heuristics matter in coordinating multiple robots under uncertainty)	5	PPT	Fish Bowl
20	MRV & Degree heuristics: choosing robot tasks with least available options first; selecting positions with maximum	5	PPT	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
	constraints (collision avoidance, shared corridors, obstacle-dense zones). LCV heuristic: assigning moves/slots that leave maximum flexibility for other robots (minimizing conflicts, maximizing cooperative navigation)			---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	Discussion on local search in CSP for multi-robot systems	5	PPT	--- NOT APPLICABLE ---
20	Min-Conflicts heuristic explained with robot navigation example. Modeling a multi-robot navigation CSP with conflicting constraints (collision avoidance, shared corridors, task scheduling)	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 13

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	Introduction to scheduling/timetabling as CSP for multi-robot systems	5	PPT	--- NOT APPLICABLE ---
20	MRV heuristic: choosing tasks with the fewest available slots first (robots with tight deadlines, limited charging stations, or restricted corridors); LCV heuristic: assigning slots that leave maximum flexibility for other robots	5	PPT	Brain Storming
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	SAT-intuition: modeling multi-robot navigation as SAT variables (moves), clauses (constraints); Encoding collision avoidance, shared resources, sensor uncertainties)	5	PPT	--- NOT APPLICABLE ---
20	Explainability: Why a constraint failed. Model a multi-robot navigation scenario as SAT, then analyze constraint failures to explain infeasible paths (e.g., two robots competing for the same corridor or charging station)	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	Discussion on stochastic environments in Robo Navigation (uncertain movements, obstacles, sensor noise)	5	PPT	--- NOT APPLICABLE ---
20	Deterministic vs stochastic decisions using robot movement example (grid world navigation) Expectimax algorithm with Robo Navigation (decision nodes = robot moves, chance nodes = uncertainty)	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 16

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	Discussion on Bounded Rationality in Robo Navigation (limited battery, partial map, time constraints)	5	PPT	--- NOT APPLICABLE ---
20	Policy Selection in robot navigation (optimal path, reward-based movement, shortest vs safest path). Multi-Agent Reasoning (multiple robots / dynamic obstacles / traffic-like coordination)	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 17

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
20	Introduction to Reasoning Under Uncertainty in Robo Navigation: Concept of uncertainty in evidence-based decision making, role of probability in handling uncertain environments	5	PPT	--- NOT APPLICABLE ---
20	Bayes' Rule and Bayesian Networks for Probabilistic Reasoning in Robot Navigation. Updating probability of obstacle detection using Bayes' Rule, Probabilistic Decision Making for Path Selection in Robot Navigation: Selection of safe paths based on probability values, comparison of deterministic and probabilistic approaches	5	PPT	Think / Pair / Share
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
20	Inference by Variable Elimination: Identification and removal of irrelevant variables, simplification of probabilistic computations for decision making	5	PPT	--- NOT APPLICABLE ---
20	Belief Propagation for State Estimation in Robo Navigation: Sampling-Based Inference Methods for Probabilistic Estimation: Introduction to rejection sampling and likelihood weighting, use of random sampling for approximate inference in uncertain environments	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 19

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	PPT	--- NOT APPLICABLE ---
20	Robot navigation under uncertainty, hidden states and multiple weaker data sources	5	Talk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
20	Motivation to Markov chains, HMM intuition for tracking to handle hidden states of the environment sensor fusion reasoning – Generate Robot Sensor fusion	5	PPT	Case Study
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
20	Robot navigation for Utility-Based Decisions in Path Planning	5	PPT	--- NOT APPLICABLE ---
20	Introduction to Utility-Based Decision Making under Uncertainty Expected Utility for Action Selection in (multi) Robot movement decision making	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 21

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap	1	Talk	--- NOT APPLICABLE ---
20	Build a hybrid reasoning model that integrates rule-based navigation with search and constraint filtering for a grid environment. They will document reasoning steps showing how rules and constraints reduce the number of explored paths.	5	PPT	--- NOT APPLICABLE ---
20	An AI reasoning pipeline integrates multiple components such as state-space representation, data structures, search algorithms, and heuristic reasoning to process inputs and generate intelligent decisions. In Python, lists, dictionaries, sets, and graph structures are used to represent knowledge, store intermediate states, and manage reasoning steps for solving complex navigation tasks.	5	Talk	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	One minute paper

SESSION NUMBER : 22

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap	1	Talk	--- NOT APPLICABLE ---
20	An intelligent agent for robot navigation is designed with a structure consisting of sensors to perceive the environment, a processing unit (agent program) to make decisions, and actuators to move the robot toward the goal while avoiding obstacles;	5	Talk	--- NOT APPLICABLE ---
20	The concept of the PEAS model is used to design a robot navigation agent by defining Performance (reach the goal safely and quickly), Environment (grid with obstacles), Actuators (wheels and motors), and Sensors (camera and distance sensors) to describe how the agent interacts with its surroundings. This concept helps in clearly understanding agent behavior and decision-making, ensuring the robot senses environmental inputs and performs appropriate actions to successfully achieve its navigation goal.	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 23

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap	5	Talk	--- NOT APPLICABLE ---
20	Design a probabilistic decision model where robots choose paths based on predicted risks and success probabilities. They will compare deterministic and probabilistic decision outputs and analyze accuracy differences.	5	PPT	--- NOT APPLICABLE ---
20	Probabilistic inference allows AI systems to make predictions under uncertainty by calculating likelihoods and probabilities of outcomes. Decision logic integrates probability values with utility-based reasoning to select actions that maximize expected success	5	Talk	One minute paper
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 24

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance & Recap	1	Talk	--- NOT APPLICABLE ---
20	ethics and risk assessment project analyzing fairness, reliability, and safety issues in AI-based navigation systems.	1	Talk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

20	Ethical AI design requires identifying potential bias in heuristics, unfair decision logic, and unreliable probabilistic predictions that may affect outcomes. Understanding these limitations prepares learners to apply data-driven learning methods used in modern AI systems.	5	PPT	Case Study
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

Tutorial Course DELIVERY Plan:

NO Delivery Plan Exists

Tutorial Session wise Teaching – Learning Plan

No Session Plans Exists

Practical Course DELIVERY Plan:

NO Delivery Plan Exists

Practical Session wise Teaching – Learning Plan

No Session Plans Exists

Skilling Course DELIVERY Plan:

Skilling session no	Topics/Experiments	CO-Mapping
1	To develop the intelligent agent robot programs in PyCharm, implement basic programs like Variables and Data Types- Variables Used to store robot position and goal. Lists (Used for Grid Environment - 2D Lists, Indexing, Nested lists)	CO1
2	To implement Graph-Based Knowledge Representation – (Nodes, Edges, dictionary) and Rule-Based Knowledge Representation.	CO1
3	To implement intelligent agent program using variables, lists, if-else, loops, functions, and random.	CO1
4	To develop a path using lists and store node costs using dictionaries to analyze grid navigation easily.	CO1
5	To implement and compare BFS, DFS, and UCS to find and evaluate optimal paths for robot navigation in grid environments.	CO2
6	To Design and Implement Greedy Best-First Search and A Algorithms for Robot Path Navigation in Grid Environments Using Heuristic Functions	CO2
7	To implement A* and IDA* with admissible heuristics to navigate obstacle-based grids efficiently and compare their memory usage.	CO2

Skilling session no	Topics/Experiments	CO-Mapping
8	To design experiments and analyze the performance of search algorithms on grid and state space problems.	CO2
9	to develop robot navigation as a CSP and use backtracking with constraint techniques to find efficient paths.	CO3
10	CSP modelling; backtracking; constraint propagation	CO3
11	CSP heuristics (MRV, Degree, LCV) to efficiently solve multi-robot navigation problems under uncertainty.	CO3
12	compare local search and backtracking methods and use the Min-Conflicts heuristic for multi-robot navigation CSPs.	CO3
13	to implement model and optimize multi-robot scheduling using CSP with MRV and LCV heuristics.	CO4
14	model multi-robot navigation as a SAT problem and analyze constraint failures for collision avoidance and explainability.	CO4
15	To design and implement a robot navigation system using the Expectimax algorithm for decision-making under uncertainty.	CO4
16	To analyze bounded rationality, policy selection, and multi-agent reasoning using Robo Navigation scenarios.	CO4
17	To implement Markov Chains and HMMs in Robot navigation under uncertainty scenario.	CO5
18	Students able to understand and implement Sensor fusion techniques in Robot navigation under uncertainty scenario.	CO5
19	understand and implement utility based decision making techniques in Robot navigation.	CO5
20	compare different utility based decision making techniques in Robot navigation.	CO5
21	to identify and analyze ethical issues such as bias, fairness, and safety in AI-based navigation systems.	CO6
22	To design a framework for evaluating reliability and risk in intelligent decision-making processes.	CO6
23	To implement bias detection and risk assessment mechanisms within an AI navigation model.	CO6
24	To evaluate system performance and propose improvements for building fair, reliable, and safe AI systems	CO6

Skilling Session wise Teaching – Learning Plan

SESSION NUMBER : 1

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	4	Talk	--- NOT APPLICABLE ---
20	To develop the intelligent agent robot programs in PyCharm, implement basic programs like Variables and Data Types- Variables Used to store robot position and goal. Lists (Used for Grid Environment - 2D Lists, Indexing, Nested lists)	4	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	4	PPT	--- NOT APPLICABLE ---
20	To implement If-Else Conditions (Decision Making), Loops (While Loop - Used for robot movement), Functions, Tuples (Used for Positions), use basic Libraries (Only small libraries needed).	4	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	4	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	4	Talk	--- NOT APPLICABLE ---
20	To design the robot navigation environment with dynamic elements (like a grid environment that includes, Start position (S), Goal position (G), Obstacles (X), Dynamic obstacle (D) → moves randomly, Free path (0) and display the grid.	4	Talk	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	4	PPT	--- NOT APPLICABLE ---
20	To implement Graph-Based Knowledge Representation – (Nodes, Edges, dictionary) and Rule-Based Knowledge Representation.	4	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	4	Talk	--- NOT APPLICABLE ---
5	Conclusion and Summary	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	4	Talk	--- NOT APPLICABLE ---
20	To design a robo grid navigation problem using state-space representation, Implement simple straight-line movement from the start state to the goal state using basic loops and coordinate updates.	4	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	4	Talk	--- NOT APPLICABLE ---
20	To implement user-controlled navigation, where movements are defined step-by-step using actions like up, down, left, and right while tracking transitions between states	4	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	4	Talk	--- NOT APPLICABLE ---
5	Conclusion and Summary	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	4	Talk	--- NOT APPLICABLE ---
20	To implement grid navigation path tracking using Python lists: Track the sequential movement of an agent in a grid environment using a list, where each visited cell is appended to store the complete navigation path from start to goal efficiently.	4	Talk	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	4	Talk	--- NOT APPLICABLE ---
20	To implement node-to-cost mappings in grid navigation using Python dictionaries : Associate each grid cell (node) with a cost value using a dictionary, where the cost represents the number of steps taken from the start state to reach each node, enabling simple evaluation of path lengths.	4	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	4	Talk	--- NOT APPLICABLE ---
5	Conclusion and Summary	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	To Implement and Compare BFS, DFS, and UCS for Robot Navigation in Grid Environments	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	To Design and Evaluate Optimal Pathfinding Algorithms for Mobile Robot Navigation System	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	Design and Implement Greedy Best-First Search and A Algorithms for Robot Path Navigation in Grid Environments Using Heuristic Functions	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	Talk	--- NOT APPLICABLE ---
20	To Implement and Compare Heuristic-Based Robot Path Navigation Using Greedy Best-First Search	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	Talk	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
20	To Design and Implement A Robot Navigation with Admissible Heuristics and Obstacle Handling in Grid Environments	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	Talk	--- NOT APPLICABLE ---
20	To Implement and Compare A and IDA Algorithms for Memory-Efficient Robot Path Navigation in Obstacle-Based Grids	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	Talk	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	Design Experiments and Perform Empirical Profiling of Search Algorithms in Grid-Based Robot Navigation and Puzzle State Spaces.	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	To Implement and Compare Search Algorithm Performance Using Graph-Based and Implicit State Space Problems Through Experimental Analysis	5	Talk	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	To Model Robot Navigation in Grid Environments as a Constraint Satisfaction Problem Using Backtracking and Constraint Propagation Techniques	5	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	To Design and Implement CSP-Based Robot Path Navigation with Forward Checking and Arc Consistency for Efficient Pathfinding	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
20	To Model Robot Navigation in Grid Environments as a Constraint Satisfaction Problem Using Backtracking and Constraint Propagation Techniques	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	To Design and Implement CSP-Based Robot Path Navigation with Forward Checking and Arc Consistency for Efficient Pathfinding	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
20	To Design and Apply Heuristic-Based CSP Techniques (MRV, Degree, LCV) for Efficient Multi-Robot Navigation in Grid Environments	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	Talk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
20	To Implement and Evaluate Heuristic Strategies for Constraint Satisfaction in Coordinated Multi-Robot Path Planning Under Uncertainty	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
20	To Design and Compare Local Search and Backtracking Methods for Multi-Robot Navigation in Constraint-Based Environments.	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	To Implement Min-Conflicts Heuristic for Solving Multi-Robot Navigation CSP with Collision and Resource Constraints	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 13

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
20	To Model Multi-Robot Scheduling and Timetabling as a Constraint Satisfaction Problem Using MRV and LCV Heuristics .	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	To Design and Optimize Task Allocation in Multi-Robot Systems Using CSP-Based Scheduling with MRV and LCV Strategies	5	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
20	To Model Multi-Robot Navigation as a SAT Problem with Constraint Encoding and Collision Avoidance.	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	To Design and Analyze SAT-Based Multi-Robot Navigation with Conflict Diagnosis and Explainability of Constraint Failures	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
20	To Model Multi-Robot Navigation as a SAT Problem with Constraint Encoding and Collision Avoidance.	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	To Design and Analyze SAT-Based Multi-Robot Navigation with Conflict Diagnosis and Explainability of Constraint Failures	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 16

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
10	Robot navigation system project discussion & Environment Setup	5	PPT	--- NOT APPLICABLE ---
20	Implement bounded rational decision-making by applying constraints such as limited depth, time, or resources to select a feasible (satisficing) navigation path instead of an optimal one.	5	PPT	--- NOT APPLICABLE ---
25	Design and compare navigation policies while implementing basic multi-agent coordination to enable multiple robots to make efficient decisions and avoid collisions in a shared grid environment.	5	PPT	--- NOT APPLICABLE ---
25	Testing & Analysis	5	PPT	--- NOT APPLICABLE ---
10	Viva	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 17

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
20	Predicting the next state of the robot using Markov chain	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	Working a robot under a partially observable state using Hidden Markov models	5	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
20	Robot navigating a warehouse using noisy sensors How do they collaborate safely – using sensor fusion	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	Introducing multi-agents under uncertain conditions like noisy sensors	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 19

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
20	Robot navigating a warehouse with uncertain obstacle information and risky path choices — selecting actions safely using utility-based decision making.	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	Multiple robots navigating a shared space with uncertain intentions and collision risks, using utility-based decision making to achieve safe and coordinated behavior.	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
20	Robot navigating a warehouse with certain Vs uncertain, reward Vs utility, speed Vs safety comparisons.	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	Multiple robots navigating a shared space with uncertain sensor data and collision risks reward Vs utility, speed Vs safety comparisons.	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 21

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance and Recap	5	Talk	--- NOT APPLICABLE ---
20	To Design and Implement a Hybrid AI Reasoning Architecture Integrating Knowledge Representation, Constraint Satisfaction, and Heuristic Search for Optimal Grid-Based Navigation.	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	To Develop an Advanced Intelligent Navigation Framework Combining State-Space Modeling, Multi-Strategy Search, and Adaptive Decision-Making for Complex Pathfinding Environments	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 22

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	To Design and Evaluate a Probabilistic Decision Model for Risk-Aware Robot Navigation with Comparative Analysis of Deterministic and Probabilistic Strategies	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	Talk	--- NOT APPLICABLE ---
20	To Develop a Decision-Theoretic Navigation Framework Using Probabilistic Inference and Utility-Based Reasoning for Optimized Path Selection Under Uncertainty	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	Talk	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 23

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	To Design and Conduct a Comprehensive Performance Analysis Framework for Comparing Algorithm Behavior Across Diverse Navigation Scenarios with Explainable Decision Tracing	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	To Develop an Explainable AI-Based Evaluation System for Analysing Search Algorithm Efficiency, Identifying Bottlenecks, and Optimizing Navigation Performance Through Failure Analysis	5	Talk	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	Talk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 24

No Session Outcomes are mapped

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	5	Talk	--- NOT APPLICABLE ---
20	To Design and Implement an Ethical AI Navigation Framework for Assessing Fairness, Reliability, and Safety in Decision-Making Systems	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	PPT	--- NOT APPLICABLE ---
20	To Develop and Evaluate a Risk-Aware Intelligent Navigation System Incorporating Bias Detection, Reliability Analysis, and Safe Decision Logic for Responsible AI Deployment	5	PPT	--- NOT APPLICABLE ---
25	EXECUTION, Analysis and viva	5	Talk	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Talk	--- NOT APPLICABLE ---

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co
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COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	H-S2,H-S5,H-S6,H-S9	---	H-S7,H-S8	H-S1,H-S4	H-S1,H-S4	H-S3	H-S3	-- -	-- -
	Tutorial	--	---	--	--	--	--	--	-- -	-- -
	Lab	--	---	--	--	--	--	--	-- -	-- -
	Skilling	--	---	--	--	--	--	--	-- -	-- -
Tue	Theory	H-S8	H-S2	H-S5	H-S6,H-S9	--	H-S7	--	-- -	-- -

	Hour	1	2	3	4	5	6	7	8	9
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	H-S1,H-S4	--	H-S2	--	H-S6,H-S9	--	H-S7	--	--
Wed	Theory	--	---	---	--	--	--	--	--	--
	Tutorial	--	---	---	--	--	--	--	--	--
	Lab	--	---	---	--	--	--	--	--	--
	Skilling	H-S1,H-S3	---	---	H-S2,H-S7	H-S2,H-S7	H-S6,H-S9	H-S6,H-S9	--	--
Thu	Theory	--	--	---	--	--	--	--	--	--
	Tutorial	--	--	---	--	--	--	--	--	--
	Lab	--	--	---	--	--	--	--	--	--
	Skilling	H-S2,H-S4,H-S7,H-S9	H-S3	---	H-S5,H-S8	H-S5,H-S8	H-S1	H-S1	--	--
Fri	Theory	---	---	--	--	--	--	--	--	--
	Tutorial	---	---	--	--	--	--	--	--	--
	Lab	---	---	--	--	--	--	--	--	--
	Skilling	---	---	H-S6	H-S3	H-S3	H-S4,H-S5,H-S8	H-S4,H-S5,H-S8	--	--
Sat	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--
Sun	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--

REMEDIAL CLASSES:

Supplement course handout, which may perhaps include special lectures and discussions that would be planned, and schedule notified according

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCs
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DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCs
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EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Hours)	CO1	CO2	CO3	CO4	CO5	CO6
End Semester Summative Evaluation Total= 40 %	End Semester Exam	Weightage	28		180	4	4	4	4	6	6
		Max Marks	100			15	15	15	15	20	20
	SEM End Project	Weightage	12		180	2	2	2	2	2	2
		Max Marks	120			20	20	20	20	20	20
In Semester Formative Evaluation Total= 24 %	Continuous Evaluation - Project	Weightage	12		120	2	2	2	2	2	2
		Max Marks	120			20	20	20	20	20	20
	ALM	Weightage	12		120	2	2	2	2	2	2
		Max Marks	60			10	10	10	10	10	10
In Semester Summative Evaluation Total= 36 %	Skill In-Sem Exam	Weightage	12		120	6	6				
		Max Marks	50			25	25				
	Semester in Exam-II	Weightage	12		120			6	6		
		Max Marks	50					25	25		
	Semester in Exam-I	Weightage	12		120	6	6				
		Max Marks	50			25	25				

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course. In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments.

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Supplement course handout, which may perhaps include special lectures and discussions

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Supplement course handout, which may perhaps include special lectures and discussions

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Sandeep Chitreddy	L	1-MA,9-MA	-	-	-	-
Sandeep Chitreddy	S	1-MA,9-MA	-	-	-	-
SUKLA SATAPATHY	L	5-MA,7-MA	-	-	-	-
SUKLA SATAPATHY	S	5-MA,7-MA	-	-	-	-
Harsha P	L	6-MA,4-MA	-	-	-	-
Harsha P	S	4-MA,6-MA	-	-	-	-
Sudarshan Yerragunta	L	8-MA,3-MA,2-MA	-	-	-	-
Sudarshan Yerragunta	S	2-MA,3-MA,8-MA	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes and carry the text book(s) or material(s) as prescribed by the Course Faculty to the class.

NOTICES

Most of the notices are available on the LMS platform.

All notices will be communicated through the institution email.

All notices concerning the course will be displayed on the respective Notice Boards.

Signature of COURSE COORDINATOR

(Sudarshan Yerragunta)(9935)(AI&DS)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of AI&DS

HEAD OF DEPARTMENT:

Approval from: DEAN-ACADEMICS

(Sign with Office Seal)

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